FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

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CSC577

SOFTWARE ENGINEERING THEORY AND PRINCIPLES

PROJECT TITLE :

SpeakOut

SpeakOut - Anonymized Bullying Reporting and Support Platform

GROUP :

NBCS2305A - Group B

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1. Introduction

1.1 Overview

SpeakOut is a safe online reporting system that will help students and school communities to report an incident of bullying with safety and anonymity. Empowering victims and witnesses to share their voices without fear of being exposed, SpeakOut gives the space in which bullying can be addressed, without any blame-shifting and with great transparency.

The website has an easy to use system to post anonymous reports where the user can post evidence in the form of photos, videos or documents. Each report comes up with a case ID that enables reporters to monitor the progress of their case as it happens.

To the school authorities, SpeakOut offers an all-encompassing case management dashboard to effectively investigate, triage and dismiss reports. Such sophisticated services like TF-IDF analysis are used to reveal the pattern and prioritize cases to improve response time and the general safety at school.

1.2 Problem Statement

When it comes to bullying, the victims tend to conceal it and do not report the cases of bullying. One of the major reasons why the victim or a bystander does not report in relation to bullying is the fear of being retaliated by the bully. It becomes more threatening when the offender is the influential or popular individual in the group of peers as it may result in fearing the loss of social position by the rest. This leads to the other factor that causes students to fear taking action that will further deteriorate the situation.

In addition, it has been discovered that students will become less inclined to report subsequent incidents in case they do not receive an appropriate response to their prior reports or that the response did not result in the successful interventions. That is why it is essential to develop a reporting system, which will be confidential and has enough support that would give students an opportunity to report about the cases of bullying without being threatened.

1.3 Assumptions about Existing Process

The assumption is that the majority of schools nowadays use manual or verbal modes of reporting the cases of bullying like students going directly to teachers or counselors. In these processes, lack of anonymity and confidentiality make students unwilling to report. Reporting in schools is normally done by hand in the form of paper-based reports or simple spread sheets, which may lead to inefficiencies and problems in record keeping. The assumption is also that the internet connection is present in the school, and the students and staff have access to the most basic digital gadgets like a smartphone, a tablet, or a computer.

Moreover, the cases reported should be reviewed and handled only by the school staff who have the right to do it like counselors or discipline teachers. The schools are also assumed to have in place existing policies or procedures to deal with the incidents of bullying after they are reported. Finally, the assumption that there are likely to be numerous students who are not entirely aware of their rights and channels they can use to report bullying leads to continued underreporting.

1.4 Motivation

The purpose of the project is to develop a secure, user-friendly and data-driven solution that will facilitate the awareness of anti-bullying. By being scalable and applicable in educational and entrepreneurial objectives, it fits the mapping objectives. It wil be addressing a major social issue that negatively affects the welfare of the students, their performance at school and their psychological growth and in severe circumstances, they self-harm or even commit suicide.

Bullying has remained a serious issue in schools, which in most cases results in emotional damage, declining academic output, and long-term psychological impacts among students. Sadly, a good number of incidents go unreported because of the fear of being retaliated against or feeling social pressure or simply because of not having confidence in the available reporting systems. This scenario highlights the need to have a safe, private, and open reporting protocol that allows the students to express their experiences in a manner that is free of apprehension and stigma. The SpeakOut platform is meant to solve this problem through a digital solution to encourage anonymous reporting and effective management of cases. With the help of the automated and transparent system of reporting, which will replace the traditional one, SpeakOut will help to achieve safer schools and promote the culture of openness and responsibility.

As an entrepreneur, the project has a high possibility of general application and commercialization. The same system can be modified to use in universities, workplaces or other organizations that want to use confidential ways to report misconduct or harassment. By adding artificial intelligence (AI), it is also possible to add value to the system through automating repetitive administrative processes, analyzing text-based reports with TF-IDF methods, and creating meaningful insights into the trends of bullying. Such AI-based functions allow prioritizing cases more rapidly, based on the data, and more suitable intervention options, which makes SpeakOut unlike traditional reporting systems.

2. Objectives and Scope

2.1 Objectives

The SpeakOut project redefines precise quantifiable objectives that drive growth and give a benchmark on what success is. These goals are SMART (Specific, Measurable, Achievable, Relevant and Time-bound) goals to make the project execution realistic and focused.

1. **Development Objective 1:** To create an anonymized bullying reporting system, which allows students to send reports about incidents safely with the ability to upload evidence, track the status in real-time, and fully cover themselves with end-to-end encryption and anonymization procedures.
2. **Development Objective 2:** To design and deploy an intelligent case management system giving school authorities (teachers, counselors, administrators) an opportunity to track, investigate, and resolve cases of bullying systematically with automated notifications, status updates and workflow transparency that generate trust and accountability.
3. **Development Objective 3:** To incorporate AI/ML-based analytics, which involve TF-IDF feature extraction and machine-based classification models to automatically determine the severity level of reports, trends, and patterns of bullying, and according to the information, offer proactive intervention strategies.

Evaluation Objective: To test the utility, precision, and applicability of the created system via pilot testing of partner schools, measure the accuracy of AI classification (target [?]85%), user satisfaction scores of the system among students and authorities, performance indicators of the system (page load time <3 seconds), and the effectiveness of the system in resolving cases using feedback surveys and usage analytics.

Such goals are directly related to system requirements and testing. The functional and non-functional requirements in each development objective will be outlined in the Software Requirements Specification (SRS) and the success criteria in the evaluation objective will be quantifiable and will be used to set the testing plan and validation efforts that will be documented in the Software Test Document (STD).

2.2 Scope

This scope definition gives a clear understanding of what the project will and will not entail, and as such there will be no over-extension of the project and yet it will be realistic and within the timeframe and available resources.

In Scope *(Included)*	**Out of Scope *(Not Included)***
iOS and Android (React and React Native) Web and mobile applications.	In-house AI/ML model training (will use pre-trained models instead)
Secure account creation and identity protection Anonymous reporting system with secure account creation and identity protection.	Live chat and video counseling with the counselor and students.
Functionalities of uploading evidence (images, videos, and papers) up to 50MB per file.	Integration with police or legal systems (first version concentrates on resolving it at school level)
Status monitoring in real-time with case IDs.	Robotized disciplinary measures (platform makes suggestions; decision-making is still with the authorities)
The automated system of notification through email and push notifications.	Social media content moderation of social media platforms beyond the internal reporting.
Elaborate dashboard to authorities inclusive of case management, investigation tools and resolution workflows.	Parent portal providing access to direct report on children (in the future)
Combination of AI/ML based on TF-IDF feature extraction and severity classification based on pre-trained Scikit-learn models.	Multi-language assistance (Pilot phase: English only)
Trends and hotspots Inference based on clustering algorithms (K-means, DBSCAN) to detect patterns.	Hardware or IoT integration (surveillance cameras, panic buttons, and so on)

Guidelines, helplines, support information, Safety resource center.	Rewards system or gamification features reporting.
Simple analytics, basic reporting, having statistics and visuals (charts, graphs, heat maps).	Mental health service provider network/directory network (enhanced in the future) of third-party mental health service providers.
RESTful API endpoints integrating with school portals and management systems	Real-time video evidence streaming
Role based access control (students, teachers, counselors, administrators) and user authentication.	Verification system or cryptocurrency payments, based on blockchain.
Encryption and security processes on the data handled with PDPA-compliance.	Pilot phase large-scale deployment or nationwide commercialization testing.
Testing and validation piloting in partner schools.	Voice recognition or speech recognition.

3. Requirements Analysis

The purpose of this section is to translate the development objectives of the proposed AI-Based Bullying Reporting and Support System into measurable and testable requirements. This ensures that the intended functions and quality targets of the system can be validated through well-defined evaluation criteria.

3.1 Development Objective

Our project goal is to create a bullying report and support system with AI capabilities to allow anonymous reporting, classify the forms of bullying based on TF-IDF, and give advice to victims and authorities with the help of GenAI. The following are some of the things that our system intends to do:

1. Offer secure, sensitive, and easy-to-use reporting experience among students and parents.
2. Identify and categorize prevalent types of bullying which include cyber, verbal, physical, and social exclusion using the text classification of TF-IDF.
3. Early intervention facilitated by automatically sending reports to the appropriate authority (teacher, counselor or education officer).
4. Provide understanding and safety support via a chatbot that is powered by GenAI.
5. Create value to school management through an administrative dashboard.

3.2 Functional Requirements (FR)

The following table demonstrates the key functional behaviors the system needs to perform:

ID	Functional Requirement	Description
FR1	The system will enable users to report bullying instances on an anonymous basis using a chatbot.	Allows secure, confidential reporting.
FR2	The system shall accept text and optional attachments (images/screenshots) for bullying evidence.	Supports multimedia input.
FR3	The system will categorize the reports based on the types of bullying (e.g., cyber, verbal, physical) with the help of TF-IDF-based text analysis.	Automates content categorization.
FR4	The system shall provide empathetic responses and safety guidance through a GenAI chatbot.	Improves user support and comfort.
FR5	The system shall route verified cases to appropriate authority dashboards (e.g., counselor or school officer).	Supports case management.
FR6	The system shall notify users about report status updates anonymously.	Ensures follow-up transparency.
FR7	The system shall allow authorized staff to review, tag, and update case status.	Enables administrative action.

3.3 Non-Functional Requirements (NFR)

The non-functional requirements explain the quality attributes and operational standards expected of the system.

ID	Non-Functional Requirement	Type	Target / Metric
NFR1	The system shall process and classify each report within 5 seconds.	Performance	p90 ≤ 5s
NFR2	The system shall maintain at least 95% uptime.	Reliability	≥ 95%
NFR3	The chatbot shall achieve at least 80% relevance in AI-generated responses (based on test feedback).	Output Quality	≥ 80%
NFR4	The interface shall be mobile-friendly and accessible to students with minimal technical skills.	Usability	SUS ≥ 70
NFR5	The system shall anonymize all user data and comply with PDPA standards.	Security	100% compliance
NFR6	The platform shall support concurrent access from at least 100 active users.	Scalability	Load test target
NFR7	All stored data shall be encrypted in transit and at rest.	Security	Encryption validated

3.4 Evaluation Objectives and Testing Plan

Each non-functional requirement is linked to an Evaluation Objective (EO), which defines how the system's performance, quality, and user experience will be verified.

EO Theme	Related NFRs	Test / Metric	Acceptance Criteria
EO-P (Performance & Reliability)	NFR1, NFR2, NFR6	JMeter load test, uptime monitor	p90 ≤ 5s, uptime ≥95%
EO-Q (Output Quality)	NFR3	Human evaluation on 50 test cases	≥80% relevance
EO-UX (User Experience)	NFR4	SUS survey + task completion	SUS ≥70, sat ≥4/5
EO-S (Security & Compliance)	NFR5, NFR7	Code review, PDPA audit checklist	100% compliance

3.5 Traceability Matrix

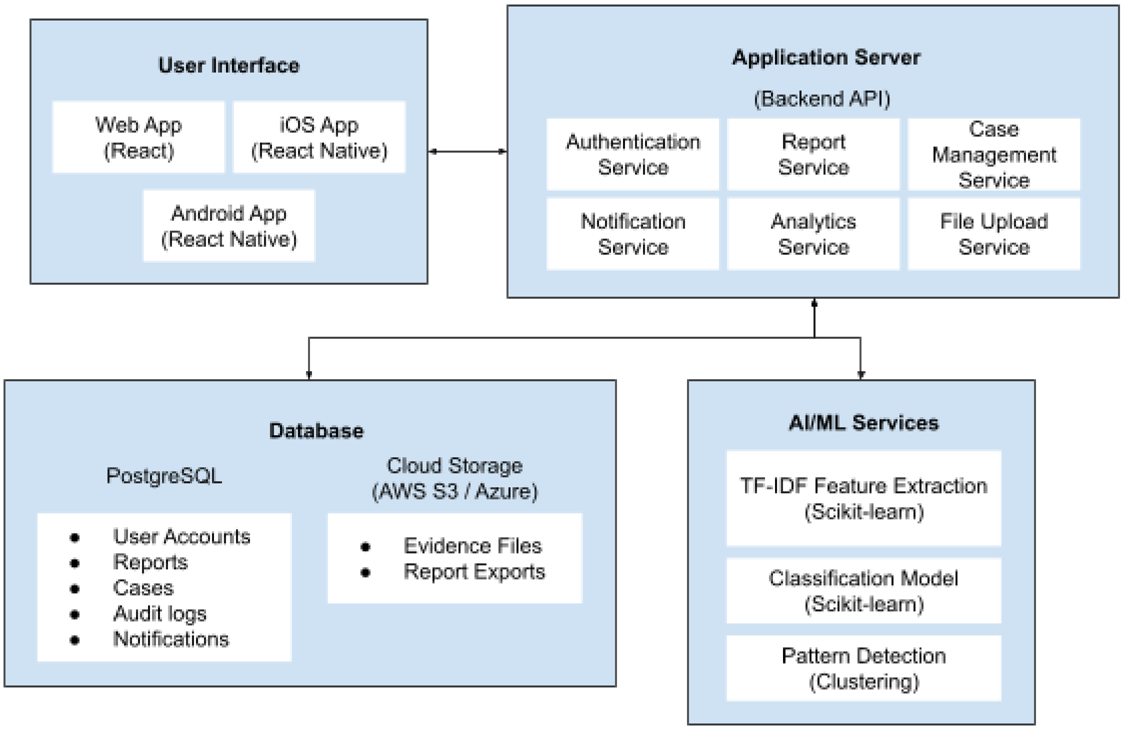
The following table links the project's development objectives to its functional and non-functional requirements, ensuring all goals are measurable and verifiable.

Development Objective	FRs	NFRs	EO Theme
Anonymous bullying reporting	FR1, FR2	NFR4, NFR5	EO-UX, EO-S
Text classification with TF-IDF	FR3	NFR1, NFR2	EO-P
GenAI-assisted chatbot support	FR4	NFR3, NFR4	EO-Q, EO-UX
Alert and case routing system	FR5, FR6, FR7	NFR2, NFR6	EO-P

4. System Design Overview

4.1 High-Level System Design

The SpeakOut platform follows a four-tier architecture consisting of the User Interface layer, application server layer, database layer and AI/ML services component.



4.2 Component Interactions

1. **User Interaction Flow:**
   - Students use the platform in both web applications and mobile applications.
   - Provide text description of reports and evidence files.
   - Get status updates on a real-time basis through notifications.
2. **Report Processing Flow:**
   - Report Service receives submissions and gives a case ID number.
   - The evidence is placed in a secure cloud storage in File Upload Service.
   - AI/ML Services evaluate report content sentimentally and severely.
   - Notification Service notifies responsible persons.
3. **Case Management Flow:**
   - Case Management Service is accessed by authorities using dashboard.
   - Record case status and place notes on investigation.
   - Analytics Service uses data to determine patterns.

- Reporters are notified back about the system.

4. **Data Security Flow:**
    - Auth Service verifies every request using JWT.
    - Sensitive information is encrypted in the database.
    - Communications are all encrypted in HTTPS/TLS.
    - All access and manipulation is followed in audit logs.

4.3 AI Integration Strategy

SpeakOut platform combines AI/ML functionalities without model training by utilizing pre-trained models and AI APIs running on clouds:

1. **Emotional State Assessment Sentiment Analysis**
    - **Technology**: TF-IDF (Term Frequency-Inverse Document Frequency vectorization with Scikit-learn
    - **Purpose:** Evaluate the tone of emotions in the incident report to determine the psychological condition and urgency of the reporter.
    - **Integration Method:**
        - Post report text messages to TF-IDF to convert text to numerical feature vectors.
        - Receive sentiment results (positive, negative, neutral) and emotional (fear, anger, sadness) indicator.
        - Apply these results to give priorities to cases that need urgent action.
    - **Implementation Method:** Python with the openai or transformers library.
2. **Severity Classification**
    - **Technology:** TF-IDF text to Python text classification models trained on Scikit-learn using TF-IDF.
    - **Purpose:** Classify the reports automatically into levels of severity (low, medium, high, critical) depending on the content and context.
    - **Integration Method:**
        - Use pre-trained Scikit-learn models that were trained using labeled datasets of bullying incidents (in academic repositories).
        - Use TF-IDF to extract features.
        - Use algorithms of classification (Random Forest, SVM, or Logistic Regression).
        - Receive severity predictions with scores of confidence.
    - **Implementation Method:** Loaded into Flask with joblib or pickle

3. **Pattern Detection and Trend Analysis**
    - **Technology:** Clustering algorithms from Scikit-learn.
    - **Purpose:** Determine patterns in the events of bullying like frequency of location, time, or profile of victims to facilitate prevention initiatives.
    - **Integration Method:**
        - Summarize report data at the temporal, spatial and categorical levels.
        - Use clustering algorithms to cluster the similar incidents.
        - Heat map and trend charts for graphics visualization
        - Produce automatic insights and recommendations to authorities.
    - **Implementation Method:** Python scripts on scikit-learn, pandas and matplotlib /plotly to visualize.
4. **Keywords Extraction using Natural Language Processing**
    - **Technology:** spaCy / NLTK for named entity recognition and keyword extraction combined with TF-IDF
    - **Purpose:** Retrieve important information out of the report text that is not structured (locations, involved parties, type of incidents).
    - **Integration method:**
        - NLP pipeline process report.
        - Mining auto entities and keywords using TF-IDF.
        - Fill in organized fields of the database.
        - Enhance the search and filtering functions.
    - **Implementation Method:** Pre-trained spaCy models (encoreweb_sm) embedded into the Report Service.
5. **Intervention Strategies Recommendation System**
    - **Technology:** Expert system that uses rule-based and collaborative filtering.
    - **Purpose:** Recommend suitable measures to the authorities to implement as per the case history results.
    - **Integration Method:**
        - Determine regulations depending on the characteristics of the cases (severity, type, history).
        - Compare existing cases to past similar cases.
        - Suggest solutions that have resulted in effective solutions.
    - **Implementation Method:** Python custom rule engine with case database reference.

4.4 Integration Architecture

- All AI products are included in a distinct microservice layer.
- Flask backend is used to communicate with the AI services with internal APIs.
- Rate limiting on API to control costs and performance.
- Redundant AI API calls prevention through caching.
- Contingencies plans in case of a failure in AI services.

4.5 Pretrained Model Approach has the following advantages

- No requirements in the form of a large labeled training data or computational power.
- Reduced time to market, tested reliable models.
- New models update regularly, and become better.
- Pay-per-use API pricing models which are cost effective.
- Put an emphasis on integration and user experience, not on model development.

5. Software Entrepreneurship Considerations

SpeakOut is aimed at primary and secondary schools that desire to enhance safety of students. Bullying is common to many students, but not reported. The UNICEF Malaysia (2023) research discovered that one in every ten students was bullied. This indicates that there is a great demand in a secure and confidential reporting system. The emergence of online technology in education suggests that the likelihood of schools using SpeakOut is also a possibility.

SpeakOut allows students to report bullying without concerns of name and without fear. It provides the schools with real-time tracking of cases and identification of bullying cases. It makes sure that it takes less time to act in response and have knowledge of the data, unlike paper reports or complaint boxes. SpeakOut can help schools to create a more safe, trusting and transparent atmosphere.

SpeakOut is able to operate on a Software-as-a-Service (SaaS) platform. In their CSR, companies have the opportunity to sponsor schools. Afterward, the system can be extended to universities or workplaces to receive reports of harassment. The model promotes sustainability and long-term growth.

6. Risk Planning

Risk	Description	Mitigation Strategy
Invasion of Data Privacy and Anonymity.	Bullying reports, personal stories, or tracking data, which are extremely sensitive, may be disclosed, and the trust in the user will be lost.	Application of anonymize or hash stored data.
Malicious or False Report	Users post fake reports in order to waste the time of the staff or free wrong people, which may interfere with investigations.	Apply TF-IDF (Term Frequency-Inverse Document Frequency) to categorize the textual data and indicate the reports where there are some keywords or patterns typical of malicious/spam data.
Discrimination in Artificial Intelligence Triage or Classification.	The results of the analysis provided by the AI/ML Models are culturally biased or they favor certain type of reports, which compromises the investigation process.	Enforce a content filtering and content review process in order to have ethical and proper handling of cases. This system should be backed by a wide variety of training data, which is ethically collected and has a human in the loop mechanism of concluding the cases.
API Failure (System Outage)	The inability to submit urgent reports on the case by students or view the case management dashboard by administrators is due to API	Implement a retry mechanism.

	downtime or cloud hosting problems.	
Regulatory & Legal Compliance	Does not adhere to the required rules that have been set by the ministry of education or laws protecting children, which may result in legal dangers.	Collaborate with the Ministry of Education and the NGOs & Child Protection Groups.

7. Team Roles and Responsibilities

Student ID	Name	Role	Job Description
2022673128	Raef Luqman Bin Mohammad Aidid	Team Leader	• Call for meeting • Coordinate tasks • Manage timelines • Work on proposal • Ensure proposal deliverable • Contribute to SRS, SDD, STD
2023862012	Anis Basyirah Binti Anas Mukhsin	Software Requirement Analyst	• Analyze and document requirements, • Communicate the requirements to the team members • Ensure SRS deliverable • Work on proposal • Contribute to SRS, SDD, STD
2023630758	Nur Aeshah Hadiera Binti Nor Azan	Software Designer and Tester	• Design software architecture and interfaces, • Develop test cases • Ensure SDD deliverable • Work on proposal • Contribute to SRS, SDD, STD
2023280798	Amrina Binti Shamsudin	Software Developer	• Ensure related technologies and libraries are working • Implement and integrate system modules • Execute tests

2023413596	Nur Dalili Husna Binti Nordin		● Ensure STR/STD deliverable ● Contribute to SRS, SDD, STD

8. Expected Deliverables

The successful completion of the SpeakOut project will result in three main outcomes which is a working software prototype with API integration, a complete set of project documentations and presentation materials. These deliverables collectively demonstrate the technical, functional and entrepreneurial viability of the solution. The following are the deliverables expected to be completed at the end of the *SpeakOut* project :

8.1 Software Prototype with API Integration

This is the main deliverable which includes a working and tested version of the SpeakOut platform. The system combines web, mobile and AI components to show how the application can be used for real-time bullying report management and awareness.

8.1.1 Web Application

A fully functional and responsive web application will be developed for both administrator and users. This version will demonstrate the complete case management process including an admin dashboard and simple data analytics.

The web application will include :

- User registration and login system with role-based access which is student (victim), PIBG (to view status report), teacher (to take action) and PDRM (to take action and have the main authority for the cases).
- Anonymous reporting form that allows students to submit reports safely with the ability to upload images and videos as evidence.
- Case management dashboard for school authorities to view, update and resolve bullying cases.
- Real-time cases tracking system that uses unique case IDs for every report.
- Notification feature to alert users about case status updates through email or in-app messages.
- Simple analytics dashboard showing data like bullying trends, report categories and response time.
- Cloud deployment so the platform can be accessed through public links.

8.1.2 Mobile Application

A mobile version of SpeakOut will be created to demonstrate how users can report bullying easily using their phones. The mobile app will focus on accessibility and usability for students, especially high school students.

The mobile application will feature :

- Built for both Android and iOS devices.
- Simple and student friendly interface with In-app help center for direct support.
- Anonymous report submission with support for text, photo and video evidence.
- Camera integration for taking and uploading images instantly.
- Push notification to update users when their report status changes.
- Offline mode allows users to draft a report even without internet access.
- Secure login system and data encryption for privacy protection.
- User feedback loop for continuous improvement of the system's interface and features.

8.2 Aligned with AI Scope and Entrepreneurship Aspects

The SpeakOut system uses TF-IDF (Term Frequency - Inverse Document Frequency) and API integration to make the application smarter and more connected. TF-IDF helps analyze the bullying reports by finding important keywords and patterns which make it easier for teachers or officers to understand each case. The API connects the web and mobile apps so data can move smoothly with real-time updates. This project also focuses on sustainability and entrepreneurship as SpeakOut can be improved and expanded for use in more schools or other institutions in the future.

The TF-IDF will feature :

- Keyword extraction which identifies key bullying-related terms from a student report.
- Report categorization helps categorize reports based on word importance and frequency for better case management.
- Pattern Analysis to detect common bullying types or frequently used negative terms to support school awareness.
- Data visualization to display keyword frequency and patterns through graphs and charts in the analytics dashboard.

API Integration :

- Connects the web and mobile applications to a central database and backend server for seamless data exchange.
- Enables communication between SpeakOut and external platforms such as school portals or government databases.
- Ensures scalability and future integration with hotline systems or other support organizations.

Key entrepreneurship aspects include :

- Problem solving innovation to address the real issue of underreported bullying in schools by providing a safe and easy to use reporting platform.
- Market scalability can be implemented in schools, universities and organizations with potential partnership with Government and NGOs.
- Business model which can adopt freemium model where basic reporting is free and advanced analytics are offered as premium features.
- sustainability and social impact to support awareness campaigns, training modules and long term collaboration to promote mental health, student safety and healthier school environment.

8.3 Project Documentations

A complete set of project documentations will be delivered to record the planning, design, implementation, and testing phases of the project ensuring future maintainability and scalability. These include :

8.3.1 Software Requirements Specification (SRS)

A detailed document outlining all functional and non-functional requirements. The SRS will contain :

- Introduction and project overview that explain the SpeakOut system, problem statement, purpose and objectives.
- Detailed functional requirements that cover user management, reporting, case management, notifications, TF-IDF text analysis and analytics.
- Non-functional requirements covering system security, performance efficiency, reliability, user friendliness, scalability, and maintainability for future improvements.
- User stories and use cases that describe the main activities and interactions of all role-based users.
- System constraint, assumptions and dependencies
- Requirements traceability matrix linking requirements to design and test cases.
- TF-IDF analysis requirements and specifications for integrating text analysis to detect key bullying keywords and recurring patterns in submitted reports.
- Acceptance criteria for each major feature before the system is approved.

8.3.2 Software Design Document (SDD)

A technical blueprint detailing the system architecture, component design, database schema, and security mechanisms. The SDD will include:

- High-level system architecture diagrams with three-tier architecture design include presentation, application and data layers.
- Backend component and module design specifications covering authentication, report management , case management, notification, analytics and file upload services.
- Detailed database schema with entity-relationship diagrams.
- API specifications and detailed endpoint documentation for managing data exchange between the web, mobile and backend services.
- User interface designs, wireframes and mockups for web and mobile applications.

- Security architecture setup including encryption mechanisms, anonymization protocols and access control.
- TF-IDF integration flow showing how reports are processed and analyzed.
- Cloud deployment architecture with infrastructure specifications for public access.
- Data flow diagrams showing how information moves through the system.

8.3.3 Software Test Report/Test Definition (STR/STD)

Documents containing the test plan, developed test cases, execution results, and defect log. The STR/STD will encompass:

- Test documentation includes test plan, developed test cases, execution results and defect log of bugs and fixes.
- Testing Approach that covers unit testing, integration testing, system testing and acceptance testing to ensure all the modules work correctly.
- Detailed test cases for all major features with expected results and actual outcomes such as user registration, report submission, notifications and case management.
- Unit test specifications for individual components and backend functions.
- Integration test scenarios to ensure APIs work correctly between web, mobile and backend systems for smooth data exchange.
- TF-IDF component testing to verify the TF-IDF analysis produces accurate keyword extraction and report categorization.
- Security and penetration test plans with vulnerability assessment results include checks for data encryption, anonymous reporting and access control.
- Usability test protocols with user feedback from pilot testing to improve user experience.
- Performance benchmarks measuring system response times, load handling and scalability to handle multiple reports at once.
- Test execution results with pass/fail status for each test case.
- Defect log documenting bugs found, severity levels, and resolution status.
- Test coverage metrics and quality assurance summary

8.4 Project Demonstration/Presentation Materials

This section focuses on the materials prepared to present and demonstrate the SpeakOut project to the examiners. The presentation highlights the project's outcomes, system functionality, social impact and potential for real world implementation. It also includes finalized documentation and supporting materials to ensure a clear understanding of the project scope and achievements. The materials include :

8.4.1 Final Report/Project Proposal Document

The complete and finalized version of this proposal and the overall project report. This will include:

- All sections of the project proposal with updates based on actual implementation.
- Lessons learned and challenges encountered during development of the system.
- Reflection on team collaboration and agile methodology application.
- Analysis of how objectives were achieved.
- Comparison of planned vs. actual deliverables.
- Future enhancement recommendations and roadmap.
- Entrepreneurship reflection on commercialization potential.

8.4.2 Final Presentation Slides

A professional presentation summarizing the project lifecycle, objectives achieved, technical approach and entrepreneurial considerations. The presentation will cover :

- Problem statement and market need with supporting statistics.
- Solution overview highlighting key features and value propositions.
- Technical architecture and TF-IDF integration strategy.
- Live system demonstration plan.
- Development methodology and team collaboration approach.
- Testing results and quality assurance outcomes.
- Impact metrics, success indicators, and pilot feedback.
- Commercialization strategy including revenue models and market analysis.
- Future roadmap and scalability plans.
- Team contributions and roles.
- Lessons learned and project reflection.
- Q&A preparation addressing potential examiner questions.

8.4.3 Live Demonstration

A compelling step-by-step walk-through of the SpeakOut prototype showcasing key features like anonymous reporting, case triage, status updates and TF-IDF analysis. The demonstration will include :

Student Reporting Workflow :

- Account creation with anonymous credentials.
- Step-by-step report submission process.
- Evidence upload demonstration such as photos, videos and documents.
- Confirmation and unique case ID generation.
- Real-time status tracking view.

Authority Case Management Workflow :

- Login to the administrator dashboard that authorized users to access the case management panel.
- View of incoming reports with TF-IDF based keyword highlights to help identify potential severity or bullying type.
- Case filtering and search functionality that allows to filter and search reports based on keywords, dates or case status for easier tracking.
- Investigation note-taking and action documentation
- Status updates triggering notifications to reporters about the progress or outcomes of the cases.
- Case resolution and closure process that are marked as resolved and the record is stored for future reference and reporting analytics.

AI Analytics and Insights :

- Real-time sentiment analysis results for submitted reports.
- Severity classification demonstration with confidence scores.
- Analytics dashboard showing bullying trends over time.
- Pattern detection with heat maps identifying hotspots.
- Automated recommendations for intervention strategies.

Mobile Application Features :

- Mobile app interface and navigation.
- Push notification demonstration.

- Cross-platform compatibility (iOS and Android).

End-to-End Scenario:

- Complete journey from report submission to case resolution.
- Demonstration of all notification touchpoints.
- Showcase of system responsiveness and performance.

8.4.4 User Guide (Basic)

A concise guide for first-time administrators and student users on how to operate the platform's core features. The user guide will provide :

For Students :

- How to create an anonymous account.
- Step-by-step instructions for submitting a bullying report.
- How to upload evidence files.
- How to track report status using case ID.
- How to access safety resources and helplines.
- Frequently asked questions about anonymity and safety.

For Administrators/Authorities :

- How to log in and navigate the dashboard.
- How to review and prioritize incoming reports.
- How to update case status and add investigation notes.
- How to assign cases to counselors or other staff.
- How to communicate with reporters anonymously.
- How to generate and interpret analytics reports.
- How to export data for official documentation.
- Troubleshooting common issues.

Technical Setup :

- Installation and deployment instructions.
- Configuration requirements.
- Database setup and initialization.
- API key configuration for TF-IDF services.
- Security best practices and maintenance guidelines.

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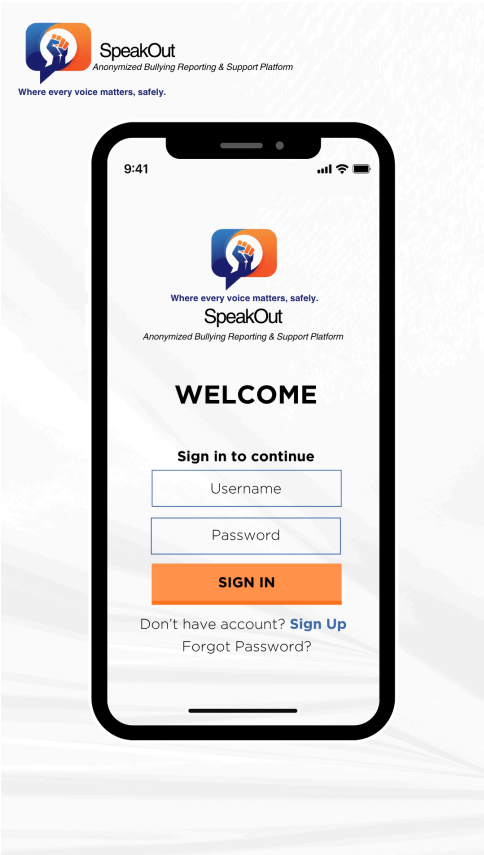
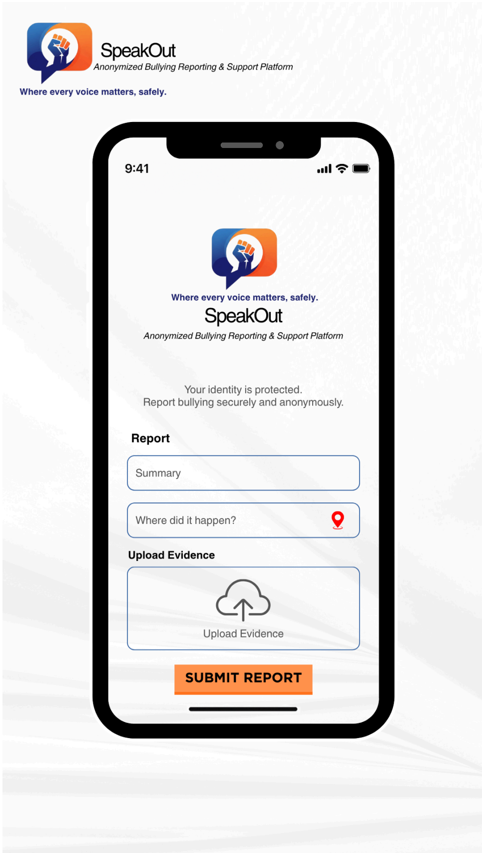
10. Appendices

Appendix A: Sample User Interface



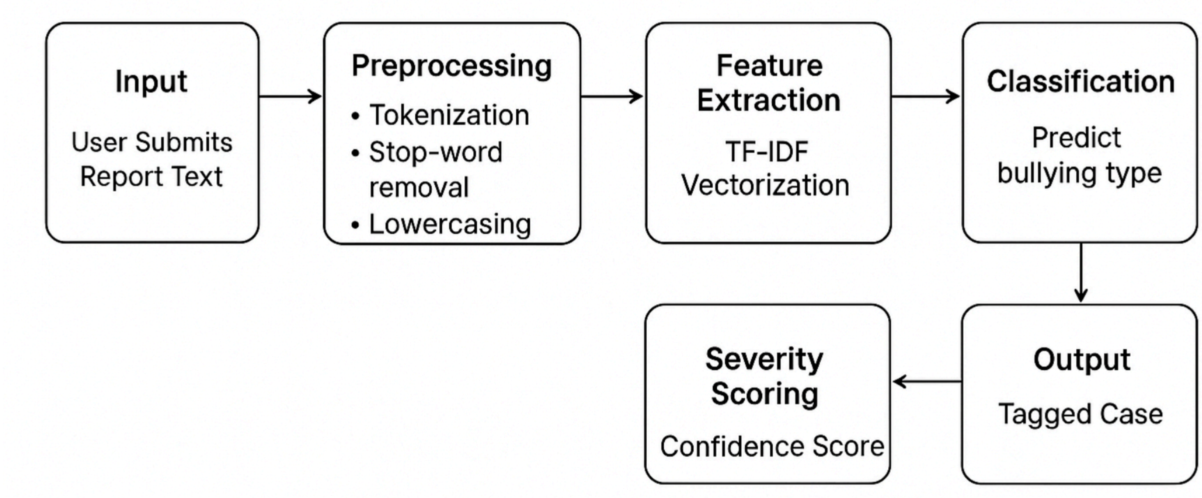


Appendix B: Extended Functional and Non-Functional Requirements

ID	Requirement Name	Description	Input/Output	Priority
FR1	Anonymous Report Submission	Allows users to submit bullying reports securely and anonymously.	Input: Text, images, videos	High
FR2	Case ID Generation	Generates a unique identifier for each report to enable tracking.	Output: Case ID	High
FR3	TF-IDF Classification Engine	Analyzes and categorizes reports based on keyword importance and context.	Input: Text	High
FR4	GenAI Response Assistant	Provides supportive replies and safety advice through chatbot.	Input: User query	Medium
FR5	Case Routing & Management	Assigns cases automatically to appropriate authorities based on severity.	Input: Classified report	High
FR6	Real-Time Notifications	Sends progress and closure updates to reporters.	Output: Notification	Medium
FR7	Authority Dashboard	Displays case data, actions, and analytics insights.	Input: Filter queries	High

ID	Requirement Name	Type	Metric	Rationale
NFR1	Processing Time	Performance	≤ 5 seconds	Reduces user frustration during report submission
NFR2	Uptime Reliability	Reliability	≥ 95%	Ensures consistent accessibility
NFR3	AI Response Accuracy	Output Quality	≥ 80%	Ensures chatbot reliability and trust
NFR4	Mobile Usability	Usability	SUS ≥ 70	Accessible for students of varying technical skills
NFR5	PDPA Compliance	Security	100% Compliance	Protects user data and identity
NFR6	System Scalability	Scalability	100 concurrent users	Ensures readiness for multi-school deployment
NFR7	Data Encryption	Security	Verified Encryption	Maintains confidentiality of all submissions

Appendix C: TF-IDF Classification Workflow



Appendix D: Sample Chatbot Conversation Flow

User Input	GenAI Chatbot Response
“Saya selalu kena ejek dalam kelas. Dalam group WhatsApp kawan2 pun sama..”	“Saya minta maaf kerana awak terpaksa lalui perkara tu. Terima kasih kerana berani berkongsi. Mari kita rekodkan laporan ini dengan selamat. Semua maklumat akan kekal **anonim**.”
“Boleh ke saya hantar screenshot ni?”	“Ya, boleh. Awak boleh muat naik tangkapan skrin atau gambar sebagai bukti. Saya akan simpan dengan selamat untuk semakan oleh guru kaunselor sekolah.”
“Mereka boleh tahu ke siapa saya?”	“Tidak. Identiti awak akan dirahsiakan sepenuhnya. Hanya **ID kes** sahaja akan dipaparkan kepada pihak sekolah.”
“Apa akan jadi lepas saya buat laporan?”	“Selepas awak hantar laporan, sistem akan **menganalisis mesej** awak untuk mengenal pasti jenis buli, dan menghantarnya kepada pihak yang sesuai untuk tindakan susulan. Awak akan menerima **kemaskini status secara anonim** di sini.”